

MANUFACTURING PROCESSES-I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 marks is allotted for right answer and 1 marks is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$
 - i. The hot working of metals is carried out -
 - a) At the recrystallization temperature
 - b) Below the recrystallization temperature
 - c) Above the recrystallization temperature
 - d) At any temperature
 - ii. Mechanical property desired for successful forging is - (a) Ductility, (b) Malleability, (c) Hardness, (d) Toughness.
 - iii. In a four high rolling mill, there are four rolls out of which -
 - a) One is working roll and three are backing up rolls
 - b) Two are working rolls and two are backing up rolls
 - c) Three are working rolls and one is backing up roll
 - d) All of the four are working rolls
 - iv. Seamless tubes in mass production are manufactured by - (a) Drawing, (b) Extrusion, (c) Rolling, (d) Casting.
 - v. Blanking and punching operations can be performed simultaneously on -
 - (a) Combination die, (b) compound die, (c) progressive die, (d) simple die.
 - vi. Lathe bed is made of a) Mild Steel b) High Carbon steel c) Cast Iron d) High Speed Steel
 - vii. The angle between face and flank of a single point cutting tool is known as a) Rake Angle b) Clearance Angle c) Lip Angle d) Nose Angle
 - viii. Which is an example of orthogonal cutting? (a) Milling, (b) Turning (c) Drilling, (d) Planning.
 - ix. Which is related with offsetting the tail stock? a) Turning b) Facing c) Taper turning d) Knurling.
 - x. The allowance given for easy removal of pattern is a) Shrinkage allowance b) Wrapping Allowance c) Machining Allowance d) Draft Allowance
 - xi. Gate is provided in moulds to - (a) Feed the casting at a constant rate, (b) Give passage to gases, (c) Compensate for shrinkage, (d) Avoid cavities.
 - xii. Metal and alloys of which following metal is used for cold chamber die casting-(a) tin, (b) lead, (c) aluminium, (d) iron
 - xiii. The flame used for welding brass is - a) Oxidising b) Neutral c) Carburising d) Natural Flame
 - xiv. The welding arc is not visible to worker in a) MMAW b) GTAW c) GMAW d) SAW
 - xv. Spot welding is one of the type of a) Arc Welding b) Resistance Welding c) Pressure Welding d) Solid-Liquid State welding

GROUP-B

Answer any Five (05) questions.

2. i) What are the advantages of hot working over cold working process?
ii) With suitable diagram explain the open die forging process. 4+4
 3. i) What do you mean by rolling? Discuss 2 Hi and 3 Hi roll stand arrangement with sketches.
ii) Name two forging defects and state the reason for arising these defects. (1+5)+2
 4. i) Discuss forward extrusion and backward extrusion with neat sketches.
ii) Describe sheet metal operations (any two) with sketch. a) Blanking, b) Punching c) Drawing. 4+4
 5. i) State the desirable properties of cutting tool materials.
ii) Explain the different types of chips formed during machining with sketches. 2+6
 6. i) Differentiate between orthogonal cutting and oblique cutting.
ii) Define Cutting speed, feed and depth of cut in Machine tool.
iii) State the functions of cutting fluid. 3+3+2
 7. i) Explain the nomenclature of a single point cutting tool with a neat sketch.
ii) Metric thread of 0.8 mm pitch is to be cut on a lathe. Pitch of the lead screw is 1.5 mm. Find the change gear ratio. 6+2
 8. i) Draw and label the elements of a gating system and briefly explain the function of each.
ii) Explain the causes and remedies of three casting defects with suitable sketches. 5+3
 9. i) Describe the different types of pattern allowances in foundry work.
ii) State the difference between MIG welding and TIG welding. 5+3
 10. i) Describe the characteristics and applications of different gas welding flames with neat sketches.
ii) Distinguish between brazing and soldering. 6+2
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